

Aufgabe 2: Nmap

- a) Wie viele Hosts befinden sich in ihrem lokalen Klasse-C-Netz?
-sn Ping scan prüft nur, welche Hosts online sind, ohne Ports zu scannen.

```
nmap -sn 192.168.0.0/24
```

```
Starting Nmap 7.96 ( https://nmap.org ) at 2025-07-08 21:52 Mitteleuropäische Sommerzeit
Nmap scan report for kabelbox (192.168.0.1)
Host is up (0.012s latency).
MAC Address: 02:10:18:B1:1C:44 (Unknown)
Nmap scan report for HF-LPx30 (192.168.0.29)
Host is up (0.0030s latency).
MAC Address: 34:EA:E7:91:57:9E (Shanghai High-Flying Electronics Technology)
Nmap scan report for DESKTOP-GCPJ12F (192.168.0.84)
Host is up (0.0100s latency).
MAC Address: A8:A1:59:B8:46:5E (ASRock Incorporation)
Nmap scan report for 192.168.0.131
Host is up (0.0040s latency).
MAC Address: A2:14:8A:7A:D4:87 (Unknown)
Nmap scan report for LGwebOSTV (192.168.0.173)
Host is up (0.015s latency).
MAC Address: F8:B9:5A:68:E3:06 (LG Innotek)
Nmap scan report for DESKTOP-0G6J098 (192.168.0.217)
Host is up.
Nmap done: 256 IP addresses (6 hosts up) scanned in 42.21 seconds
```

256 Adressen wurden gescannt, 6 davon haben geantwortet.

In Wireshark

arp						
No.	Time	Source	Destination	Protocol	Length	Info
2496	129.224095	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.173
2502	130.257832	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.173
2503	130.257832	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.21? Tell 192.168.0.173
2504	130.257832	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.191? Tell 192.168.0.173
2505	131.282327	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.191? Tell 192.168.0.173
2506	131.282327	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.21? Tell 192.168.0.173
2507	131.282327	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.173
2510	132.096593	MS-NLB-PhysServer-1	Intel_a0:00:b4	ARP	60	Who has 192.168.0.21? Tell 192.168.0.1
2511	132.096627	Intel_a0:00:b4	MS-NLB-PhysServer-1	ARP	42	192.168.0.217 is at 00:28:f8:a0:00:b4
2548	144.284315	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.191? Tell 192.168.0.173
2549	144.284315	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.21? Tell 192.168.0.173
2550	144.285252	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.173
2552	145.318323	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.173
2553	145.318323	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.21? Tell 192.168.0.173
2554	145.318323	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.191? Tell 192.168.0.173
2557	146.231990	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.191? Tell 192.168.0.173
2558	146.231990	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.21? Tell 192.168.0.173
2559	146.231990	LGInnoteK_68:e3:06	Broadcast	ARP	42	Who has 192.168.0.104? Tell 192.168.0.173
2560	147.565278	a2:14:8a:7a:d4:87	Broadcast	ARP	42	Gratuitous ARP for 192.168.0.131 (Reply)

Frame 2511: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{F8245424-00FE-4520-0000-000000000000} Ethernet II, Src: Intel_a0:00:b4 (00:28:f8:a0:00:b4), Dst: MS-NLB-PhysServer-16_18:b1:1c:44 (02:10:18:b1:1c:44)

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: Intel_a0:00:b4 (00:28:f8:a0:00:b4)

Sender IP address: 192.168.0.217

Target MAC address: MS-NLB-PhysServer-16_18:b1:1c:44 (02:10:18:b1:1c:44)

Target IP address: 192.168.0.1

0000 02 10 18 b1 1c 44 00 28 f8 a0 00 b4 08 06 00 01D (.....
0010 08 00 04 00 02 00 28 f8 a0 00 b4 c0 a8 00 d9P (.....
0020 02 10 18 b1 1c 44 c0 a8 00 01D.....

- b) Welches Betriebssystem wird von scanme.nmap.org verwendet?
-O aktiviert die Erkennung des Betriebssystems

```
nmap -O scanme.nmap.org

Starting Nmap 7.96 ( https://nmap.org ) at 2025-07-08 22:16 Mitteleuropäische Sommerzeit
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.18s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite
Aggressive OS guesses: Linux 4.19 - 5.15 (97%), Linux 4.15 (93%), IPFire 2.27 (Linux 5.15 - 6.1) (92%), Linux 5.4 (91%), Linux 3.10 (90%), Linux 2.6.32 (90%), Linux 5.0 - 5.14 (90%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (90%), Linux 3.11 - 4.9 (90%), Linux 3.2 - 3.8 (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 23 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.13 seconds
```

Der Host scanme.nmap.org verwendet ein Linux-Betriebssystem.

- c) An welchem Datum wurde die Webseite nmap.org registriert?

```
C:\Windows\System32\cmd.exe
D:\Desktop\whois>whois nmap.org

Whois v1.21 - Domain information lookup
Copyright (C) 2005-2019 Mark Russinovich
Sysinternals - www.sysinternals.com

Connecting to ORG.whois-servers.net...

WHOIS Server: whois.dynadot.com
Registrar URL: http://www.dynadot.com
Updated Date: 2023-08-31T05:05:15Z
Creation Date: 1999-01-18T05:00:00Z
Registry Expiry Date: 2029-01-18T05:00:00Z
Registrar: Dynadot Inc
Registrar IANA ID: 472
Registrar Abuse Contact Email: abuse@dynadot.com
Registrar Abuse Contact Phone: +1.6502620100
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Registry Registrant ID: REDACTED
Registrant Name: REDACTED
Registrant Organization: Super Privacy Service LTD c/o Dynadot
Registrant Street: REDACTED
Registrant City: REDACTED
Registrant State/Province: California
Registrant Postal Code: REDACTED
Registrant Country: US
Registrant Phone: REDACTED
Registrant Phone Ext: REDACTED
Registrant Fax: REDACTED
Registrant Fax Ext: REDACTED
```

Creation date is 18.01.1999.

- d) Wie kann man möglichst effektiv eine größere Menge an Adressen nach offenen TCP-Ports scannen?

In dieser Teilaufgabe benutze ich Befehl `nmap -sS -p 1-1000 -T4 192.168.0.0/24`.

Dabei steht `-sS` für einen SYN-Scan, der nur den ersten Schritt der TCP-Verbindung ausführt und somit besonders schnell und unauffällig ist. Mit `-p 1-1000` werden die wichtigsten Standardports gescannt, und `-T4` (Timing Level 4) macht einen aggressiven, schnellen Scan und reduziert Wartezeiten.

```
nmap -sS -p 1-1000 -T4 192.168.0.0/24
```

Starting Nmap 7.96 (<https://nmap.org>) at 2025-07-08 22:25 Mitteleuropäische Sommerzeit
Warning: 192.168.0.29 giving up on port because retransmission cap hit (6).
Warning: 192.168.0.131 giving up on port because retransmission cap hit (6).
Warning: 192.168.0.173 giving up on port because retransmission cap hit (6).
Nmap scan report for kabelbox (192.168.0.1)
Host is up (0.0077s latency).
Not shown: 994 closed tcp ports (reset)

PORT	STATE	SERVICE
22/tcp	filtered	ssh
23/tcp	filtered	telnet
53/tcp	open	domain
80/tcp	open	http
111/tcp	filtered	rpcbind
443/tcp	filtered	https

MAC Address: 02:10:18:B1:1C:44 (Unknown)

Nmap scan report for HF-LPx30 (192.168.0.29)
Host is up (0.0038s latency).
All 1000 scanned ports on HF-LPx30 (192.168.0.29) are in ignored states.
Not shown: 960 closed tcp ports (reset), 40 filtered tcp ports (no-response)
MAC Address: 34:EA:E7:91:57:9E (Shanghai High-Flying Electronics Technology)

Nmap scan report for A80-von-Karlo (192.168.0.31)
Host is up (0.022s latency).
All 1000 scanned ports on A80-von-Karlo (192.168.0.31) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 46:CD:37:C5:AB:9C (Unknown)

Nmap scan report for DESKTOP-GCPJ12F (192.168.0.84)
Host is up (0.011s latency).
All 1000 scanned ports on DESKTOP-GCPJ12F (192.168.0.84) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: A8:A1:59:B8:46:5E (ASRock Incorporation)

Nmap scan report for 192.168.0.131
Host is up (0.0074s latency).
All 1000 scanned ports on 192.168.0.131 are in ignored states.
Not shown: 860 closed tcp ports (reset), 140 filtered tcp ports (no-response)
MAC Address: A2:14:8A:7A:D4:87 (Unknown)

Nmap scan report for Redmi-Note-13 (192.168.0.135)
Host is up (0.024s latency).
All 1000 scanned ports on Redmi-Note-13 (192.168.0.135) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 1A:3A:03:88:E2:24 (Unknown)

Nmap scan report for LGwebOSTV (192.168.0.173)
Host is up (0.0063s latency).
All 1000 scanned ports on LGwebOSTV (192.168.0.173) are in ignored states.
Not shown: 953 closed tcp ports (reset), 47 filtered tcp ports (no-response)
MAC Address: F8:B9:5A:68:E3:06 (LG Innotek)

Nmap scan report for DESKTOP-0G6J098 (192.168.0.217)
Host is up (0.0011s latency).
Not shown: 996 closed tcp ports (reset)

PORT	STATE	SERVICE
135/tcp	open	msrpc
137/tcp	filtered	netbios-ns
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds

Nmap done: 256 IP addresses (8 hosts up) scanned in 147.02 seconds

e) Wie funktioniert der SYN-Scan und für was kann man ihn verwenden?

Beim SYN-Scan (-sS) sendet Nmap nur ein SYN-Paket an einen Port. Wenn der Port offen ist, antwortet der Zielhost mit SYN-ACK, aber Nmap die Verbindung nicht fortsetzt, sondern mit einem RST-Paket abbricht.

Dieser Scan ist besonders schnell und unauffällig, da keine vollständige Verbindung aufgebaut wird und somit keine Einträge in normalen Logs entstehen.

SYN-Scans eignen sich gut zum schnellen Auffinden offener TCP-Ports in lokalen und entfernten Netzwerken, auch hinter einfachen Firewalls.

f) Welches sind die offenen Ports, die bei Ihren bisherigen Nmap-Scans am häufigsten auftreten, und wofür werden sie verwendet?

53/tcp	open	domain
80/tcp	open	http
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds

- 53 (DNS) wird für die Namensauflösung im Netzwerk, z.B. beim Zugriff auf Websites oder andere Dienste verwendet.
- 80 (HTTP) ist Webserverport für unverschlüsselte Webseiten.
- 135, 139, 445 (Windows-Dienste) werden für Datei- und Druckerfreigabe sowie Remote-Prozeduraufrufe (RPC) in Windows-Netzwerken verwendet.